

AKTIO, Greece

WMO: 166430

Lat: 38.925N

Lon: 20.765E

Elev: 3

StdP: 101.28

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.9	4.2	-7.0	2.1	6.0	-4.6	2.6	6.7	13.6	10.0	11.7	10.3	4.1	40	0.437

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 9.1	(c) 33.1	(d) 22.4	(e) 31.8	(f) 22.9	(g) 30.3	(h) 23.0	(i) 25.3	(j) 29.7	(k) 24.7	(l) 29.0	(m) 24.1	(n) 28.3	(o) 4.2	(p) 270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.1	19.0	27.7	23.2	18.0	27.2	22.9	17.6	26.9	77.5	29.6	75.0	29.2	72.6	28.3	28.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 10.5	(b) 9.2	(c) 8.1	(e) 0.7	(f) 36.6	(g) 1.4	(h) 1.5	(i) -0.3	(j) 37.7	(k) -1.1	(l) 38.6	(m) -1.9	(n) 39.4	(o) -2.8	(p) 40.5		
(a) 10.5	(b) 9.2	(c) 8.1	(e) -1.6	(f) 26.8	(g) 1.6	(h) 0.9	(i) -2.8	(j) 27.4	(k) -3.7	(l) 27.9	(m) -4.6	(n) 28.4	(o) -5.8	(p) 29.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.7	10.0	10.6	12.4	15.2	19.3	23.3	25.6	26.3	23.0	19.3	15.2	11.4		
	DBStd	6.17	2.57	2.63	2.33	2.40	2.23	2.49	1.84	1.99	2.18	2.29	2.77	2.65		
	HDD10.0	78	31	22	6	1	0	0	0	0	0	0	2	16		
	HDD18.3	1104	258	217	183	98	16	1	0	0	1	16	99	216		
	CDD10.0	2874	31	39	82	157	288	398	482	504	389	289	157	59		
	CDD18.3	856	0	0	0	3	44	149	224	246	139	47	4	0		
	CDH23.3	6687	0	0	1	16	173	1091	2064	2371	847	120	5	0		
	CDH26.7	2056	0	0	0	2	20	298	686	866	178	6	0	0		
(14) Wind	WSAvg	3.6	3.7	3.9	3.7	3.5	3.4	3.5	3.6	3.5	3.4	3.3	3.4	3.7		
(15) Precipitation	PrecAvg	900	110	103	84	59	36	13	8	13	68	101	146	158		
	PrecMax	1305	285	215	168	166	101	48	48	72	176	310	339	299		
	PrecMin	535	0	9	0	13	4	1	0	0	4	0	0	21		
	PrecStd	191	71	58	44	38	24	11	11	17	47	67	65	71		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.0	19.8	21.1	25.2	29.0	34.0	34.3	35.9	31.8	27.4	23.9	19.9		
		MCWB	13.7	13.1	14.0	16.9	20.1	21.7	21.9	21.0	21.4	19.7	18.2	15.8		
	2%	DB	16.2	17.2	19.0	22.8	26.8	31.2	32.6	33.7	30.0	25.8	22.0	18.0		
		MCWB	12.8	12.9	13.9	15.9	19.3	22.6	22.8	22.1	22.4	19.5	17.4	14.8		
	5%	DB	15.1	16.0	17.8	21.0	25.1	29.8	31.1	32.0	28.8	24.8	20.8	16.8		
		MCWB	12.0	12.5	13.6	15.6	19.3	22.6	23.2	22.9	21.9	19.2	16.7	13.9		
	10%	DB	14.0	15.0	16.7	19.8	23.9	28.1	30.0	30.8	27.2	23.5	19.2	15.3		
		MCWB	11.1	11.7	13.2	15.3	19.0	22.1	23.2	23.2	21.3	18.7	15.7	12.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.8	14.4	15.7	18.7	22.1	25.5	26.1	26.5	24.9	22.1	19.5	16.7		
		MCDB	16.9	17.2	18.5	22.7	26.1	29.8	30.6	30.4	28.9	25.2	22.6	18.7		
	2%	WB	13.4	13.5	14.9	17.3	21.0	24.2	25.2	25.5	23.8	21.1	18.0	15.2		
		MCDB	15.7	16.2	17.7	20.9	24.7	28.7	29.7	29.7	27.5	24.0	21.1	17.3		
	5%	WB	12.4	12.8	14.1	16.4	20.2	23.5	24.6	24.8	23.1	20.3	17.1	14.2		
		MCDB	14.6	15.5	16.9	19.6	23.7	27.8	28.9	29.1	26.8	23.2	19.8	16.3		
	10%	WB	11.5	12.0	13.4	15.6	19.4	22.7	24.0	24.2	22.4	19.5	16.2	13.2		
		MCDB	13.5	14.6	16.0	18.6	22.9	26.9	28.2	28.4	26.0	22.4	18.7	15.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.4	6.9	7.4	7.9	8.3	8.8	9.0	9.1	8.3	7.6	6.8	6.1		
		MCDBR	7.1	7.9	8.6	9.9	10.1	10.8	10.8	11.1	9.6	8.6	7.8	7.0		
	5% WB	MCWBR	4.7	4.9	5.1	5.3	5.1	4.7	4.9	5.1	4.6	4.5	4.4	4.7		
		MCWBR	6.5	7.4	7.8	8.8	9.2	9.7	9.6	9.7	8.7	7.8	7.1	6.1		
(40) Clear-Sky Solar Irradiance	taub	taub	0.352	0.382	0.427	0.474	0.464	0.448	0.453	0.467	0.442	0.414	0.381	0.359		
		taud	2.436	2.324	2.189	2.073	2.161	2.256	2.248	2.202	2.265	2.315	2.373	2.408		
	Ebn at Noon	Ebn at Noon	812	831	825	808	824	835	827	805	802	784	766	771		
		Edh at Noon	84	106	134	160	149	136	136	139	123	106	89	81		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.97	2.77	4.09	5.39	6.55	7.45	7.65	6.74	5.01	3.50	2.32	1.77		
	RadStd	RadStd	0.22	0.32	0.37	0.44	0.34	0.36	0.27	0.25	0.30	0.28	0.20	0.27		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.37	N/A	N/A	+1.22	N/A	N/A	N/A	-92	+146	+84
(47) Regional (8 neighbors)		+0.32	N/A	N/A	+0.87	N/A	N/A	N/A	-89	+116	+60

Nomenclature: See separate page